

Which Architectures Can Read a Relative Positional Encoding? Traversal-Based Encodings in Sparse Graph Neural Networks

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Abstract

Positional encodings are the standard remedy for the 1-Weisfeiler–Leman bound on message-passing networks, and the usual question is which signal to compute. We report that how it is injected matters more. Our vehicle is a traversal-based *relative* encoding: the sinusoidally encoded hop distance $d(u, v)$ on every node pair within r hops, anchor-free where absolute traversal encodings are not, and needing rewiring to reach a sparse architecture at all. On CLUSTER the identical signal is worth +21.3 points under GAT and nothing under GCN (−0.9). That cell is three seeds, its gain is under 2σ of the control’s seed spread and its ablations are GCN-only, so we read it as a direction rather than a measurement. A proposition explains the asymmetry: since GCN and GAT both *normalise* their aggregation, an edge-weight or attention-bias injection cancels algebraically when weighted degree is constant and features uniform, so on CSL such a model provably cannot beat chance, and does not. We also place the encoding in the hierarchy, $1\text{-WL} \subsetneq \text{T-RPE} \subseteq \text{GD-WL} \subseteq 3\text{-WL}$, on explicit witnesses, and report that our pre-registered sparse-versus-dense prediction failed in the opposite direction.

1 Introduction

Message-passing graph neural networks (MPNNs) aggregate information over local neighbourhoods, and their ability to distinguish graphs is bounded by the 1-Weisfeiler–Leman colour refinement test (Xu et al., 2019; Morris et al., 2019). The bound is not a technicality: an MPNN cannot see global symmetry, measure how far apart two nodes are, or separate graphs that agree locally but differ globally.

Positional encodings (PEs) are the standard response. They enrich node features with a structural signal computed outside the message-passing recursion, and they come in two families. *Absolute* PEs assign a vector to each node, Laplacian eigenvectors (Dwivedi et al., 2023; Dwivedi and Bresson, 2021), random-walk return probabilities (Dwivedi et al., 2022), or

distances to a set of anchors (You et al., 2019; Li et al., 2020). *Relative* PEs assign a value to each node *pair*, and are the norm in graph transformers, where a pairwise bias is added to every attention logit (Ying et al., 2021; Rampásek et al., 2022).

Traversal-based encodings have so far lived almost entirely in the absolute family: a breadth-first search from an anchor produces a per-node distance that drops straight into the feature vector, but the encoding is then a function of the anchor set, and that set falls to a heuristic whose ties are broken arbitrarily. The relative formulation has no anchors: $d(u, v)$ is a property of the pair.

The obstacle is that a relative encoding has nowhere to live in a sparse MPNN, since every edge joins nodes at distance exactly one and a function of $d(u, v)$ restricted to E is constant. Making it meaningful requires changing which pairs exchange messages: rewiring (Brüel-Gabrielsson et al., 2023; Topping et al., 2022). We take the natural construction, an edge for every pair within r hops carrying its hop distance. Rewiring is not neutral — it relieves over-squashing (Alon and Yahav, 2021) but multiplies the edge count — so a plain baseline confounds the encoding with the rewiring it requires, and we run two controls: the rewired graph with a *constant* edge feature, and the unrewired graph at $r = 1$.

Our contributions:

- **Injection decides the outcome**, in theory and in measurement. Because GCN and GAT both *normalise* their aggregation, an edge-weight or attention-bias injection cancels algebraically once weighted degree is constant and features uniform (Proposition 4), so on CSL it cannot help however expressive the encoding is, and measurably does not; outside that setting the same signal is worth +21.3 points under GAT and −0.9 under GCN (§6). Prior work reports relative encodings almost only inside dense attention, where the issue never surfaces.

- **A relative traversal encoding** (T-RPE) as the vehicle (§3), placed in the expressiveness hierarchy (§4) on explicit witnesses: above 1-WL, strengthened by shortest-path counts, below 3-WL, blind to strongly regular graphs.
- **An evaluation designed around the mechanism** (§5), contrasting a sparse long-diameter benchmark with a dense short-diameter one, on a prediction registered before training.

2 Related Work

Absolute positional encodings. Dwivedi et al. (2023) introduced Laplacian eigenvector PEs, a global spectral coordinate system: expressive, but $\mathcal{O}(|V|^3)$ to decompose and defined only up to a per-eigenvector sign (Lim et al., 2023). Random-walk PEs (Dwivedi et al., 2022) record k -step return probabilities, cheaper and sign-free. You et al. (2019) and Li et al. (2020) use distances to sampled or selected anchors, the closest prior work in signal but not in form: per-node and anchor-dependent.

Relative positional encodings. Relative position is standard in sequence models (Vaswani et al., 2017), and the graph analogue lives in graph transformers: Graphormer (Ying et al., 2021) biases every attention logit by shortest-path distance and GraphGPS (Rampášek et al., 2022) unifies such biases with local message passing, both paying $\mathcal{O}(|V|^2)$ for dense attention. Our question is whether the relative view survives in a *sparse* architecture, where the bias must ride on explicitly added edges.

Rewiring and expressiveness. Alon and Yahav (2021) identified over-squashing and Topping et al. (2022) tied it to negatively curved edges, motivating schemes that add or reweight edges. Closest to us, Brüel-Gabrielsson et al. (2023) rewire using positional information and feed the encoding as an edge attribute; our construction follows theirs, and we add the traversal signal, the expressiveness analysis, and a control separating rewiring from encoding. On expressiveness, Xu et al. (2019) and Morris et al. (2019) established the 1-WL bound and Zhang et al. (2023) define generalised-distance WL (GD-WL), showing shortest-path distance already yields a strictly stronger test. We propose no new variant, only place ours.

3 Method

Traversal. For a graph $G = (V, E)$ and a radius r , a breadth-first traversal from each $u \in V$ to depth r yields the hop distance $d(u, v)$ for every v within

r hops, and, at no extra cost, the number $\#sp(u, v)$ of shortest paths realising it, at $\mathcal{O}(|V|(|V| + |E|))$ per graph, computed once. In practice we take both from r matrix products instead: the smallest k with $(A^k)_{uv} > 0$ is $d(u, v)$, and since walks of that length are exactly shortest paths, $(A^{d(u,v)})_{uv} = \#sp(u, v)$. We verify this against a reference breadth-first search.

Rewiring and encoding. We take $E_r = \{(u, v) : 1 \leq d(u, v) \leq r\}$; at $r = 1$ this is the original graph and the encoding is constant, making $r = 1$ a built-in degenerate control rather than only a hyperparameter setting. Each rewired edge carries $\gamma(d(u, v)) \in \mathbb{R}^p$, a sinusoidal embedding of its hop distance in the sense of Vaswani et al. (2017), so nearby distances receive nearby representations, and a small MLP maps $\gamma(d)$ — optionally concatenated with $\log(1 + \#sp(u, v))$ — to the edge representation e_{uv} .

Injection. The encoding enters through two channels. The first is the natural one for each architecture: in GAT, e_{uv} is an additive bias on the attention logit of (u, v) , which is what a relative positional encoding means in the transformer sense; in GCN, which has no edge-feature channel, e_{uv} becomes a learned scalar gate $w_{uv} = \sigma(\mathbf{w}^\top e_{uv})$ used as an edge weight. The second is not optional *on vertex-transitive or uniform-feature graphs*, for reasons Proposition 4 makes precise, though §6 shows it is not universally beneficial either: each node also receives its aggregated relative-distance profile,

$$p_v = \sum_{u: 1 \leq d(u, v) \leq r} \gamma(d(u, v)),$$

$$x_v \leftarrow [x_v \parallel \text{MLP}(p_v)],$$

which is anchor-free, permutation invariant, and exactly the aggregation GD-WL performs. For the $\#sp$ variant the summand is $[\gamma(d), \log(1 + \#sp) \cdot \gamma(d)]$, keeping distance and path count *jointly*: summing them separately would discard which path counts belong to which distance. Since d and $\#sp$ are isomorphism invariants of the pair, both channels are permutation equivariant and the pooled output invariant, which we verify empirically; the obvious alternative is not, a BFS *visit-order* index changing under relabelling.

4 Expressive Power

Write $c^t(v)$ for the colour of v after t refinement rounds. 1-WL refines by $c^{t+1}(v) = \text{HASH}(c^t(v), \{\{c^t(u) : u \in N(v)\}\})$. Define SPD-

WL by

$$c^{t+1}(v) = \text{HASH}(c^t(v), \{(d(u, v), c^t(u)) : u \neq v\}),$$

with $d(u, v) = \infty$, a distinguished symbol distinct from every integer, when u and v lie in different components, and SPD-WL identically but with the pair label $(d(u, v), \#sp(u, v))$. For $r \geq \text{diam}(G)$, T-RPE realises SPD-WL; the M4 variant realises SPC-WL. For $r < \text{diam}(G)$ it realises a different, weaker test in which every pair beyond r carries the label ∞ ; every experimental claim below except CSL, run at $r = 6 = \text{diam}$, uses that truncated version.

Proposition 1 (Lower bound). SPD-WL *refines* 1-WL, *strictly*.

Refinement is immediate: the neighbours of v are exactly the pairs with $d(u, v) = 1$, so any 1-WL class is a union of SPD-WL classes. For strictness, decalin versus bicyclopentyl serves (10 nodes, 11 edges, same degree sequence), and CSL gives a second witness the experiments exercise directly.

Proposition 2 (Ceiling on strongly regular graphs). *Neither SPD-WL nor SPC-WL separates two connected strongly regular graphs with the same parameters.*

Strongly regular graphs with equal parameters have identical distance distributions by definition: adjacent pairs share λ common neighbours, non-adjacent pairs μ , and $\mu > 0$ (connectedness) forces diameter 2, so every pair carries the same label under either test. The 4×4 rook’s graph and the Shrikhande graph are both SR(16, 6, 2, 2), and neither separates them.

Against 3-WL, containment holds. 2-FWL (3-WL in the usual GNN naming) refines a colouring of *pairs* and can compute matrix products over it, so it obtains every $(A^k)_{uv}$, hence $d(u, v)$ as the smallest k with $(A^k)_{uv} > 0$ and $\#sp(u, v)$ as its value there: both pair labels are functions of the 3-WL colouring, giving SPD-WL, SPC-WL $\subseteq$ 3-WL. What we leave open is *strictness*, which the rook/Shrikhande pair cannot witness, since 3-WL is stable at the initial colouring there too — the refinement multiset of a pair depends only on (k, λ, μ) , which the two graphs share. We confirmed this by running 2-FWL on both to stabilisation.

Proposition 3 (Counting paths helps). SPC-WL *refines* SPD-WL, *strictly*.

Refinement is immediate since $(d, \#sp)$ determines d . For strictness we searched exhaustively over all graphs on $n \leq 6$ nodes: a witness exists at $n = 6$ and none at $n \leq 5$, so 6 is minimal. The two graphs

(graph6 EFz? and E1N?, edge lists in Appendix B) have 6 nodes, 8 edges, the same degree sequence and, crucially, the *same* multiset of pairwise distances $\{1^{16}, 2^{12}, 3^2\}$, differing only in how many shortest paths realise them. Hop distance alone therefore provably cannot separate them, which is exactly what the extra channel is for.

Together these place the method as $1\text{-WL} \subsetneq \text{T-RPE} \subseteq \text{GD-WL} \subseteq 3\text{-WL}$ (Zhang et al., 2023).

Expressive power is not enough: injection matters.

The propositions above concern what the *encoding* can distinguish, and say nothing about whether an architecture can read it.

Proposition 4 (Normalised aggregation cancels edge weighting). *Let the node features be uniform, let the rewired edge weights depend only on $d(u, v)$, and let every node have the same weighted degree D — as holds whenever G is vertex-transitive, but constant weighted degree is the operative hypothesis. Then a GCN layer using those weights, or a GAT layer using them as attention biases, computes an output independent of the weights.*

The proof (Appendix A) turns on both architectures normalising their aggregation to sum to one. This is not a corner case: CSL, the benchmark that most sharply motivates positional encodings, is vertex-transitive with uniform features, so a weight-only model *cannot* exceed chance there, and in our experiments it scored $10.00 \pm 0.00\%$ across 20 runs. The weighting reaches the layer only through the scalar $D = \sum_k n_k w(k)$, which normalisation divides out, whereas $p_v = \sum_k n_k(v) \gamma(k)$ is vector-valued and, the $\gamma(k)$ being linearly independent, retains the whole profile $\{n_k\}$: a sum is not renormalised, which is why that channel survives. The literature reports relative encodings almost exclusively inside dense attention, where the question does not arise; carrying one into a sparse, normalising architecture makes *how* it is injected as consequential as what it encodes.

5 Experimental Setup

Datasets. We use three benchmarks from Dwivedi et al. (2023). CSL (150 graphs, 41 nodes) is 4-regular and featureless; 1-WL sees ten identical classes, so a plain GNN can only guess at 10%. We follow the stratified 5-fold cross-validation protocol Dwivedi et al. (2023) specify for it. CLUSTER (12k graphs, ~ 117 nodes, node classification) is a six-community stochastic block model with average degree ≈ 37 and diameter 2–3; exactly one node per community carries its

class label and every other node is featureless, so the task is to propagate six seeds outwards. **CIFAR10-SP** (45k graphs, ~ 117 nodes) is sparse, long-diameter 8-nearest-neighbour superpixel graphs; every condition, controls included, discards its edge weights, so all models see an unweighted graph with the superpixel coordinate as a node feature. The last two are the dense and sparse arms of a deliberate contrast, a relative distance encoding carrying information only if distances spread out. The distance profiles we measure before training (Appendix D) qualify both arms, though: at $r = 2$ CLUSTER’s E_r is nearly complete, the $3.3\times$ edge blow-up being saturation rather than growth, while the same radius reaches only 22.8% of CIFAR10-SP’s pairs at mean diameter 8.5. The dense arm is tested with its distance structure exhausted and the sparse arm with most of it truncated away.

Conditions. **M0** is the architecture with no PE (control); **M1** adds Laplacian eigenvector PEs with sign-flip augmentation; **M2** adds random-walk PEs; **M3** is T-RPE; and **M4** adds the shortest-path-count channel. Each runs under both GCN and GAT.

A prediction registered before training. The distance profiles of the CSL classes are fixed by the dataset, so the ceiling our encoding can reach there is computable in advance. At $r = 3$ only 6 of the 10 classes have distinct profiles; at $r \geq 4$, 9 do. The two that remain tied are CSL(41, 6) and CSL(41, 16), whose distance multisets are identical, and adding shortest-path counts separates them. We therefore predict, before running anything, a ceiling of 90% for M3 and 100% for M4 on CSL, and we set $r = 6$ (its diameter). For CLUSTER and CIFAR10-SP we use $r = 2$.

The same diagnostic predicts *where* the encoding can help at all: mean diameter 2.2 on CLUSTER against 8.5 for CIFAR10-SP and 6.0 for CSL, so if the method behaves as claimed it should give little or nothing on CLUSTER and help on the other two.

Protocol. Four message-passing layers, hidden width 64, residual connections, batch normalisation, Adam at 10^{-3} . Model selection is on validation accuracy and the test set is read once, at the selected epoch, rather than averaged over the final epochs. We do not match parameter counts across conditions, and the overhead is not negligible: on an ~ 18 k-parameter backbone the PE modules add 12–14% under GCN and 35–37% under GAT, so we do not claim capacity is equalised. The **rewire-only** ablation, which runs M3’s modules at 19,975 parameters against its 20,215,

settles the question directly (Table 3).

Ablations. On CLUSTER with GCN — and, we should say up front, nowhere else — we vary the radius $r \in \{1, 2, 3\}$, the encoder, and the injection depth (Appendix C). Two variants are load-bearing. **rewire-only** is the rewired graph with a *constant* edge feature — all of M3’s connectivity, none of its positional signal — and the only condition that attributes an improvement to the encoding rather than the rewiring it requires. **weight-only** and **profile-only** isolate the two channels of §3, which Proposition 4 makes vacuous and essential respectively on CSL but says nothing about on CLUSTER, where features are genuine.

6 Results and Discussion

Expressivity. Table 2 (Appendix B) confirms each proposition on its witness. On CSL, M0 is pinned at exactly 10.00 ± 0.00 in *both* architectures: chance, zero variance over 20 runs, as the 1-WL bound requires on featureless 4-regular graphs. M3 reaches 85.50 under both backbones, agreeing to the decimal, and M4 reaches 99.50/100.00. That agreement is not a coincidence but Proposition 4 as a measurement: on CSL the weight channel cancels under both architectures, so each model is reduced to a function of the same profile p_v , and backbone-independence is what the proposition predicts. The ceilings pre-registered in §5 hold: M4 meets its 100% and M3 approaches its 90% from 4.5 points below, so the prediction bounds the outcome without fixing it. Combinatorially, SPD-WL separates 97.8% of the non-isomorphic 1-WL-colliding CSL pairs and SPC-WL all of them, the 144 SPD-WL misses being precisely the CSL(41, 6)/CSL(41, 16) block: Proposition 3 has an instance on a standard benchmark. M1 and M2 also solve CSL and M3 loses to both: the ceiling working rather than the method failing, since SPD-WL cannot exceed 90% here while LapPE and RWPE are not bound by it.

The registered prediction failed. §5 committed in advance to the encoding helping where distances are spread out (CIFAR10-SP, mean diameter 8.5) and giving little on CLUSTER, at 2.2. **The opposite happened.** Our largest gain anywhere is on CLUSTER (GAT M4 at 69.85 against an M0 control of 48.57, +21.3 points, exceeding LapPE by 4.9), while CIFAR10-SP, the predicted best case, barely moved: +2.5 and +2.1 under GCN (M3, M4), +0.5 and -0.1 under GAT. That control is itself unstable across its three seeds (± 11.74 against ± 0.22 for M4) and 21.3 is under 2σ of that spread, so we read the sign, not

		CSL	CLUSTER	CIFAR10-SP
GCN	M0 no PE (control)	10.00 $\pm$ 0.00	39.89 $\pm$ 4.01	46.62
	M1 LapPE	98.83 $\pm$ 1.91	60.00 $\pm$ 1.00	47.12
	M2 RWPE	100.00 $\pm$ 0.00	35.04 $\pm$ 2.21	47.01
	M3 T-RPE (ours)	85.50 $\pm$ 4.97	37.67 $\pm$ 4.89	49.13
	M4 T-RPE + #sp (ours)	99.50 $\pm$ 2.18	39.04 $\pm$ 6.11	48.68
GAT	M0 no PE (control)	10.00 $\pm$ 0.00	48.57 $\pm$ 11.74	59.93
	M1 LapPE	97.50 $\pm$ 3.78	64.97 $\pm$ 0.64	60.46
	M2 RWPE	99.00 $\pm$ 4.36	53.30 $\pm$ 1.87	60.74
	M3 T-RPE (ours)	85.50 $\pm$ 6.69	66.58 $\pm$ 1.43	60.45
	M4 T-RPE + #sp (ours)	100.00 $\pm$ 0.00	69.85 $\pm$ 0.22	59.80

Table 1: Test accuracy (%), mean $\pm$ std. CSL: 20 runs per cell (4 seeds $\times$ 5 stratified folds); CLUSTER: 3 seeds; CIFAR10-SP: a single run, so those cells carry no variance estimate and none is printed.

the magnitude. We report the miss rather than reframe around it, and Table 2 offers an explanation: neither CLUSTER nor CIFAR10-SP produces a *single* 1-WL collision among the sampled pairs, so expressivity was never the binding constraint and an intervention justified by it had no route to helping. That diagnostic is graph-level, though, and so supports the CIFAR10-SP half only; CLUSTER is node classification, where the binding quantity is within-graph node-colour collisions, which we did not measure. The prediction was not unlucky, it was aimed at the wrong bottleneck.

Injection, not sparsity, decides the outcome. Holding dataset and encoding fixed and changing only the backbone, CLUSTER moves -2.2 and -0.9 under GCN (M3, M4) but $+18.0$ and $+21.3$ under GAT, a 20-point swing from the backbone alone. The contrast is not that GCN can never read the encoding — on CIFAR10-SP it gains $+2.5$ and $+2.1$, its best cell in that single-seed column — but that on the one dataset where the encoding has something large to give, it gives only to GAT, which is Proposition 4’s mechanism outside the setting where we could prove it. GAT admits the encoding as an additive attention-logit bias, a genuine edge-feature channel; GCN has none, so it is compressed into a scalar gate on an aggregation that is then renormalised. CLUSTER being neither vertex-transitive nor uniformly featured, the mechanism degrades the signal rather than annihilating it. The ablation agrees: under GCN **weight-only** scores 38.84 against **profile-only** at 20.79, so the channel that survives here is the one that provably cancels on CSL, and the channel that rescued CSL is actively harmful.

Two things bound that cell. At $r = 2$ CLUSTER’s rewired graph is nearly complete, so GAT with M3 or M4 is structurally close to a small dense graph transformer with a shortest-path attention bias, and its gain over a 4-layer *sparse* baseline cannot be separated from that architectural change. And the level is un-

remarkable absolutely: 69.85 is about where a plain 16-layer GAT sits in Dwivedi et al. (2023) (70.59 at 528k parameters), while our M0 (48.57) sits well below their 4-layer GAT (57.73), so the $+21.3$ is internal to this setup rather than an absolute gain.

Attribution. Table 3 and Figure 1 (Appendix C) report the ablation, all ten variants at 40 epochs and one seed against the grid’s 60 and three, so they compare only to each other ($r = 2$ scores 36.11 here against M3’s 37.67 in Table 1). Every one is GCN/CLUSTER: nothing here controls the GAT cell above.

T-RPE at $r = 2$ exceeds **rewire-only** (the same rewired graph, constant edge feature) by 15.5 points at 1.2% fewer parameters, so the gap is not a capacity artefact — but **rewire-only** is the wrong control to stop at. The ablation also contains $r = 1$, the original graph with a constant encoding (§3), at 35.20: rewiring with no positional signal costs 14.6 points against not rewiring at all, and $r = 2$ recovers only $+0.9$ over $r = 1$. Almost all of the 15.5 points is therefore T-RPE repairing damage the rewiring did to admit it, and Table 1 agrees at its own budget, GCN/CLUSTER M3 sitting 2.2 points *below* M0: the encoding does real work, mostly undoing its own precondition. Three variants beat the pre-registered configuration we report as M3 — **layerwise** injection (39.57), a **one-hot** encoder (39.34), **weight-only** (38.84) against its 36.11 — but one-hot outscoring sinusoidal is not evidence against §3’s smoothness argument: at $r = 2$ there are two distances and nothing to interpolate, so the ablation cannot test that argument at this radius.

Cost. Preprocessing is cheaper than a Laplacian decomposition but is not the real cost (Appendix E): rewiring multiplies the edge count by 2.9–6.0 $\times$, on every epoch. The radius buys nothing *on CLUSTER* ($r = 3$ scores 30.29 against 36.11), the only dataset we swept it on (§7).

7 Limitations and Conclusion

Three limitations of the method are structural: connected strongly regular graphs are exactly what the method cannot handle (Proposition 2); it has little to say on dense, short-diameter graphs; and the radius costs edges, $2.9\text{--}6.0\times$ of them, for an accuracy benefit we never measured. The last is weaker than it looks, since the radius sweep ran only on CLUSTER, whose diameter of 2.2 means larger r provably adds nothing, while §5’s pre-registration says the opposite where distances do spread ($r = 3$ separates six CSL classes, $r \geq 4$ nine, which is why CSL runs at $r = 6$).

The rest are ours, and they land on the headline cell. Our M0 controls sit below published figures on CLUSTER and CIFAR10-SP, so comparisons here should be read between conditions and not against the literature. CLUSTER is three seeds and the +21.3 is under 2σ of the control’s own spread; every ablation is GCN/CLUSTER, so that cell has no rewiring or single-channel control; and its rewired graph at $r = 2$ is nearly complete, making the cell an architectural change as much as an encoding. CIFAR10-SP ran at a single seed *and* at $r = 2$, which reaches 22.8% of its pairs, so the long-diameter arm was under-tested as well as wrong.

One tension is ours to own. §1 divides encodings into absolute (per node) and relative (per pair) and we set out to defend the relative side, but the channel that works is p_v , a per-node vector and therefore *absolute* by that taxonomy, while the genuinely relative channel is the inert one. The relative signal became usable only once projected back to per-node, which is a result about the taxonomy as much as about the method.

Our clearest results are negative, and we report them as findings: an encoding injected as an edge weight or attention bias is provably inert under normalised aggregation (Proposition 4) and empirically scores chance, and our pre-registered prediction was wrong in the opposite direction because it assumed expressivity was the binding constraint on benchmarks where we could find no 1-WL collisions at all. What survives is one positive claim, about injection rather than encoding: the same signal worth nothing under GCN is worth +21 points under GAT. The useful question is not which structural signal to compute, but which architectures can read one at all.

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A Proof of Proposition 4

Write $w_{uv} = w(d(u, v))$ for the edge weight, $n_k(u)$ for the number of nodes at distance k from u , and $\hat{d}_u = \sum_v w_{uv}$ for the weighted degree. Then $\hat{d}_u = \sum_{k=1}^r n_k(u) w(k)$, so the weighted degree is determined by u ’s multiset of incident distances. That multiset is the same at every node when G is vertex-transitive, since an automorphism carrying u to u' is a distance-preserving bijection of V ; hence $\hat{d}_u = \sum_k n_k w(k) = D$ for all u , which is the hypothesis of the proposition and the only use made of vertex-transitivity.

GCN’s symmetric normalisation then gives $\hat{w}_{uv} = w_{uv} / \sqrt{\hat{d}_u \hat{d}_v} = w_{uv} / D$, so $\sum_u \hat{w}_{uv} = \hat{d}_v / D = 1$ for every v . With $h_u = h$ for all u , the layer returns $\sum_u \hat{w}_{uv} W h_u = W h$, independent of w . For GAT, uniform features make every attention logit at v equal up to its bias $b(d(u, v))$, and the softmax renormalises those to coefficients α_{uv} summing to one; the layer returns $\sum_u \alpha_{uv} W h_u = W h$ for any bias, this time without needing constant weighted degree. In both cases the output is a convex combination of identical vectors, so the weighting survives only through the scalar D , which normalisation divides out. $\square$

B Expressivity on witness pairs

Table 2 confirms each proposition of §4 on its witness, measured combinatorially rather than through a trained model.

The Proposition 3 witness. The minimal pair found by exhaustive search over all graphs on $n \leq 6$ nodes is, on vertex set $\{0, \dots, 5\}$,

$$G_1 : 03, 04, 05, 13, 14, 15, 23, 24 \quad (\text{EFZ?}),$$

$$G_2 : 01, 03, 05, 12, 15, 23, 24, 34 \quad (\text{E1N?}),$$

both with degree sequence $(2, 2, 3, 3, 3, 3)$ and ordered-pair distance multiset $\{1^{16}, 2^{12}, 3^2\}$. Their $(d, \#sp)$ multisets differ: G_1 has

witness pair	1-WL	SPD-WL	SPC-WL
decalin / bicyclopentyl	–	✓	✓
CSL(41,2) / CSL(41,3)	–	✓	✓
CSL(41,4) / CSL(41,5)	–	✓	✓
4×4 rook / Shrikhande SR(16,6,2,2)	–	–	–
<i>1-WL-indistinguishable pairs separated, of those not isomorphic</i>			
CSL (6480 pairs)	–	97.8%	100.0%
CLUSTER (no 1-WL collisions)	–	–	–
CIFAR10 (no 1-WL collisions)	–	–	–

Table 2: Separation of witness pairs, and the share of 1-WL-indistinguishable pairs separated on each benchmark. CSL ships 15 permuted copies per class, so isomorphic pairs are excluded from the denominator; among the 120 graphs sampled from each of CLUSTER and CIFAR10-SP (see the sampling procedure above) no same-size pair collides under 1-WL, which is a statement about that sample and not about the full 12k/45k datasets.

$$\{(1, 1)^{16}, (2, 2)^8, (2, 3)^4, (3, 4)^2\} \quad \text{against} \quad G_2\text{'s} \\ \{(1, 1)^{16}, (2, 1)^8, (2, 2)^4, (3, 2)^2\}.$$

Sampling procedure for the benchmark rows.

From each dataset we take 120 graphs at evenly spaced indices (all of CSL; the validation splits of CLUSTER and CIFAR10-SP), bucket them by $(|V|, |E|)$ since only equal-sized graphs can collide, and compare every pair within a bucket. The three signatures are computed once per graph rather than once per pair. On CSL all 120 graphs are 4-regular and land in one bucket, giving 7140 pairs, of which 660 are isomorphic copies and are excluded from the denominator.

C Ablations

Figure 1 and Table 3 give the ablation on GCN/CLUSTER discussed in §6: radius $r \in \{1, 2, 3\}$, the four distance encoders, injection depth, and the three control variants (**rewire-only**, **weight-only**, **profile-only**). All ten variants share a budget of 40 epochs at seed 0, so they are comparable to each other but not to Table 1, which was trained for 60 epochs at three seeds. There is no matching sweep under GAT, which is the control gap §6 and §7 flag: the two rows the attribution argument turns on, $r=1$ and **rewire-only**, exist for GCN only.

D Distance diagnostics

Figure 2 gives the per-dataset hop-distance histogram measured before any training. Mean diameters are 6.0 (CSL), 2.2 (CLUSTER) and 8.5 (CIFAR10-SP); CLUSTER places 100% of node pairs at distance ≤ 2 , against 28.5% for CSL and 22.8% for CIFAR10-SP. Edge blow-up $|E_r|/|E|$ at $r = 2$ is $2.9\times$ (CSL), $3.3\times$ (CLUSTER, already saturated there) and $3.3\times$

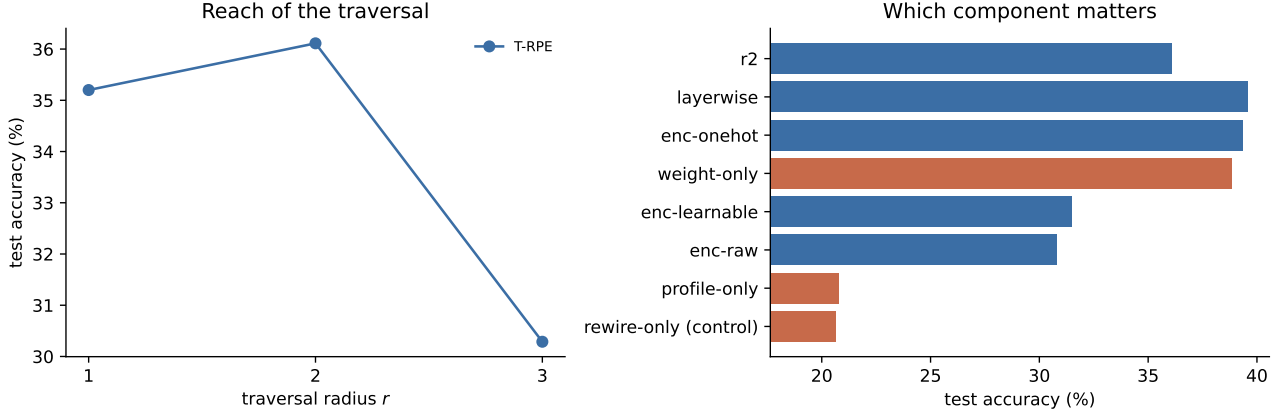


Figure 1: Ablations on GCN/CLUSTER, all variants at a matched budget of 40 epochs at one seed. **rewire-only** and the single-channel variants are highlighted. T-RPE at $r = 2$ exceeds **rewire-only** by 15.5 points, but only $r=1$ (the unwired graph with a constant encoding) by 0.9; see §6, where we read the two together.

variant	r	channels	#Param	test acc.
$r=1$ (unwired control)	1	both	20 215	35.20
$r=2$ (= M3)	2	both	20 215	36.11
$r=3$	3	both	20 215	30.29
rewire-only	2	both	19 975	20.64
raw scalar encoder	2	both	19 975	30.81
one-hot encoder	2	both	20 007	39.34
learned-table encoder	2	both	20 263	31.49
layerwise injection	2	both	21 898	39.57
weight-only	2	weight	18 615	38.84
profile-only	2	profile	20 215	20.79

Table 3: The ten ablation variants, GCN/CLUSTER, 40 epochs at seed 0. Chance is 16.7% on six-class CLUSTER. These rows compare to each other and not to Table 1, which ran a longer schedule; the same numbers are plotted in Figure 1.

(CIFAR10-SP), rising at $r = 3$ to $5.2\times$ for CSL and $6.0\times$ for CIFAR10-SP. This diagnostic is what the prediction in §5 was derived from, and it remains correct as a description of the graphs; what it failed to predict is discussed in §6.

E Cost

Figure 3 gives preprocessing cost per graph against LapPE and RWPE, together with the edge blow-up $|E_r|/|E|$. The blow-up, not the preprocessing, is the cost that recurs on every epoch.

F Training curves

Figures 4 and 5 plot every one of the 240 grid runs: validation accuracy and training loss against epoch, for all five conditions under both backbones on all three datasets. Faint lines are individual runs, bold is their mean and the band ± 1 standard deviation; mean and band are drawn only while at least half the runs

in a cell are still training, since runs stop at different epochs and averaging over the survivors past that point would show the tail of one run rather than a mean.

There is no per-epoch *test* curve because the test set is read once, at the epoch selected on validation (§5); the dotted horizontal lines in Figure 4 mark that single final test accuracy per condition.

Two things are visible here that the tables cannot show. On CSL, M0 sits flat on the 10% chance line under both backbones and stops early: the 1-WL bound as a training curve rather than a number. On CLUSTER the GCN and GAT panels separate cleanly: M3 and M4 track the baselines under GCN but pull away from them under GAT, which is the injection effect of §6 developing over training.

G Reproducibility

Every experiment is a self-contained Colab notebook; each run writes one JSON file, and all tables and figures in this paper are regenerated from those files. The repository holds the 30 leaf notebooks, the 252 run artifacts under `trpe/results/` (240 grid runs, the ten ablations, the expressivity and efficiency records), and the collection code that turns them into this paper’s tables and figures: `shared/07_results_and_figures.ipynb` for Tables 1 and 2 and Figures 1–3, `shared/10_ablation_table.py` for Table 3, and `shared/08_training_curves.py` for Figures 4 and 5. Code: https://github.com/henm2000/graphs_machine_learning.

Distance profile: how much information a relative encoding can carry

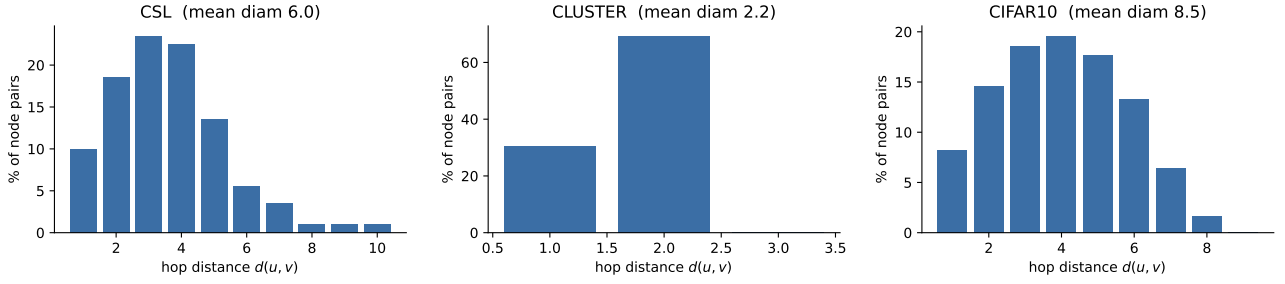


Figure 2: Hop-distance distribution per dataset, measured before training.

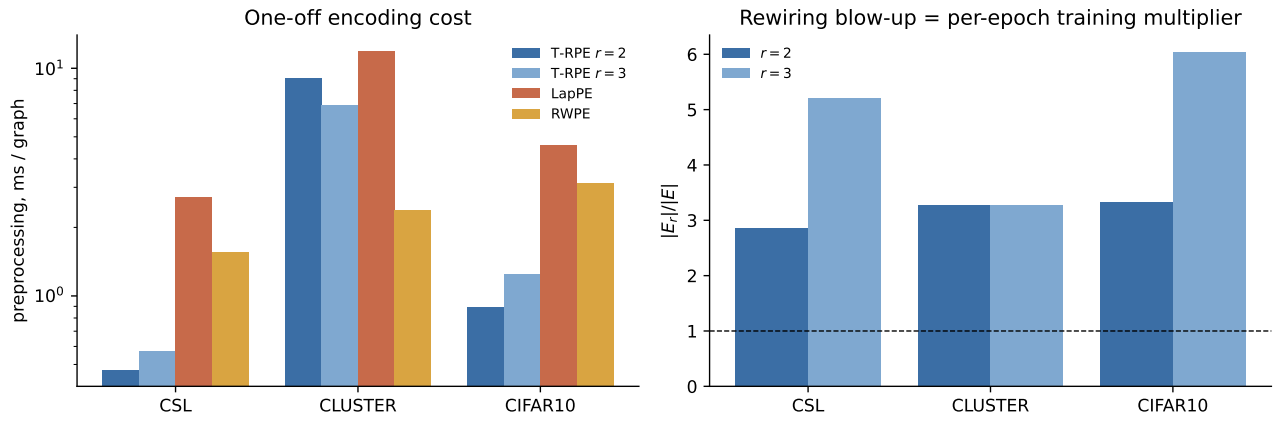


Figure 3: Preprocessing cost per graph, and the edge blow-up $|E_r|/|E|$.

Validation accuracy per epoch. Faint lines are individual runs; bold is the mean, band is ± 1 s.d.; dotted line is final test accuracy.

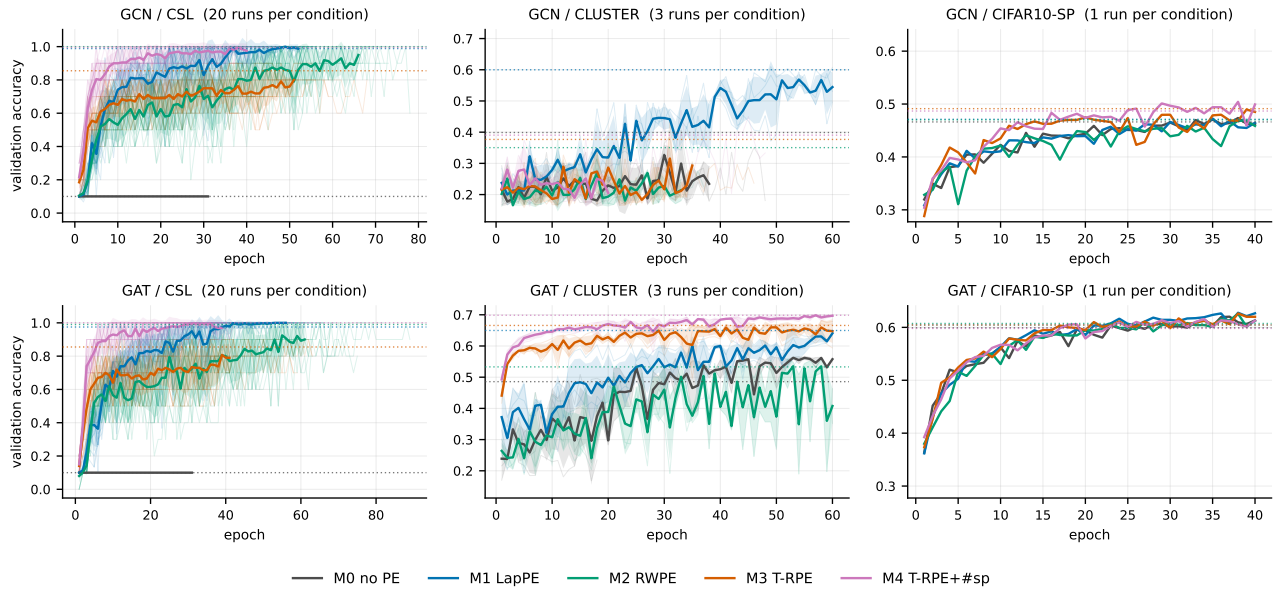


Figure 4: Validation accuracy per epoch, all 240 grid runs. Faint lines are individual runs, bold the mean, band ± 1 s.d.; dotted horizontal lines are the final test accuracy, read once at the selected epoch.

Training loss per epoch. Faint lines are individual runs; bold is the mean, band is ± 1 s.d.

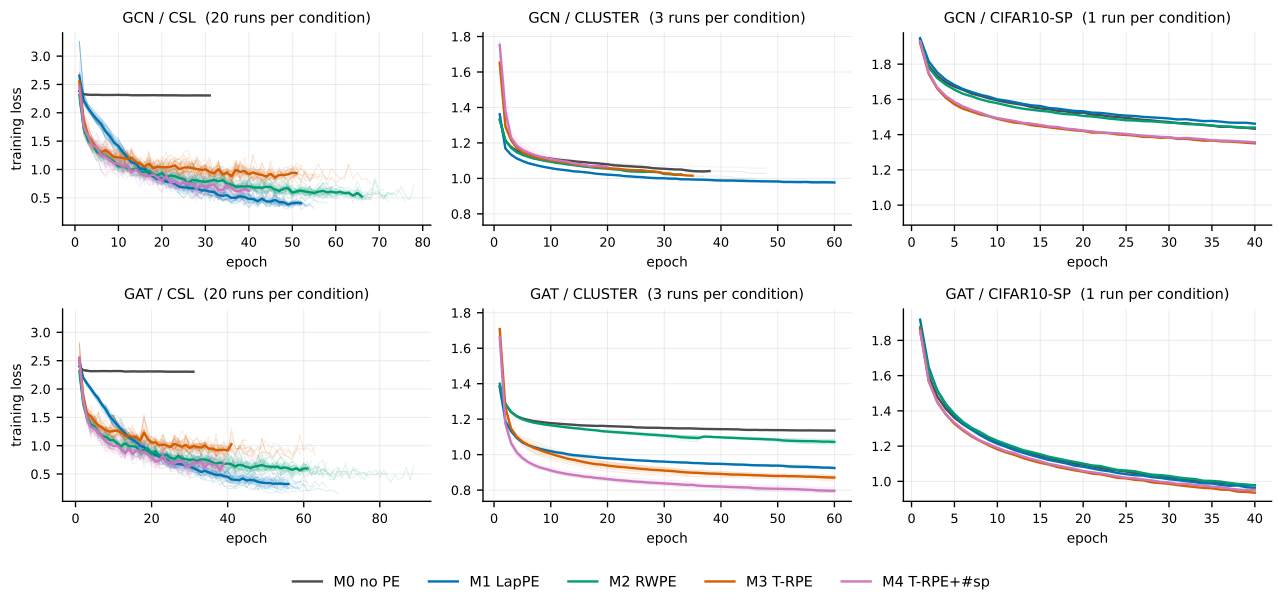


Figure 5: Training loss per epoch, all 240 grid runs. Faint lines are individual runs, bold the mean, band ± 1 s.d.